

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

APAI Lab03: DNN Shrinking and Quantization

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In this Hands-on session:

A first-time user of Pytorch framework will learn how to :

- shrink a NN, by acting on the number of layers, channels, or stride factor
- Quantize a NN down to 2 bits
- Use Netron to visualize a ONNX representation of a CNN

Tasks:

1. Load model's trained weights of LAB1;
2. Reduce network's size under 5 MMAC;
3. Re-train the reduced network and verify network's accuracy;
4. Quantize with QuantLab;
5. Export Onnx and analyze the float32 and quantized models with Netron.

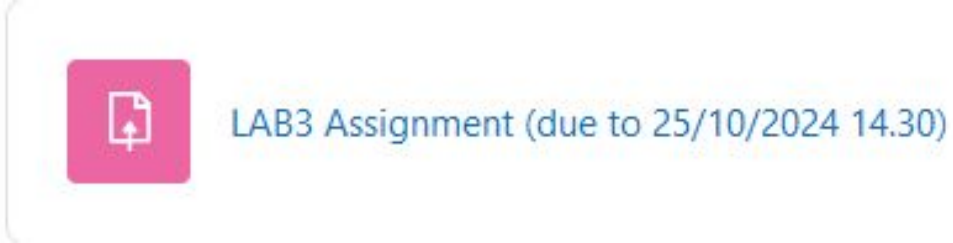
All the details about the tasks are explained in the pdf document attached.



How to deliver the assignment

- Use Virtuale platform to load your file
- update only the .ipynb file, **named as follows**:

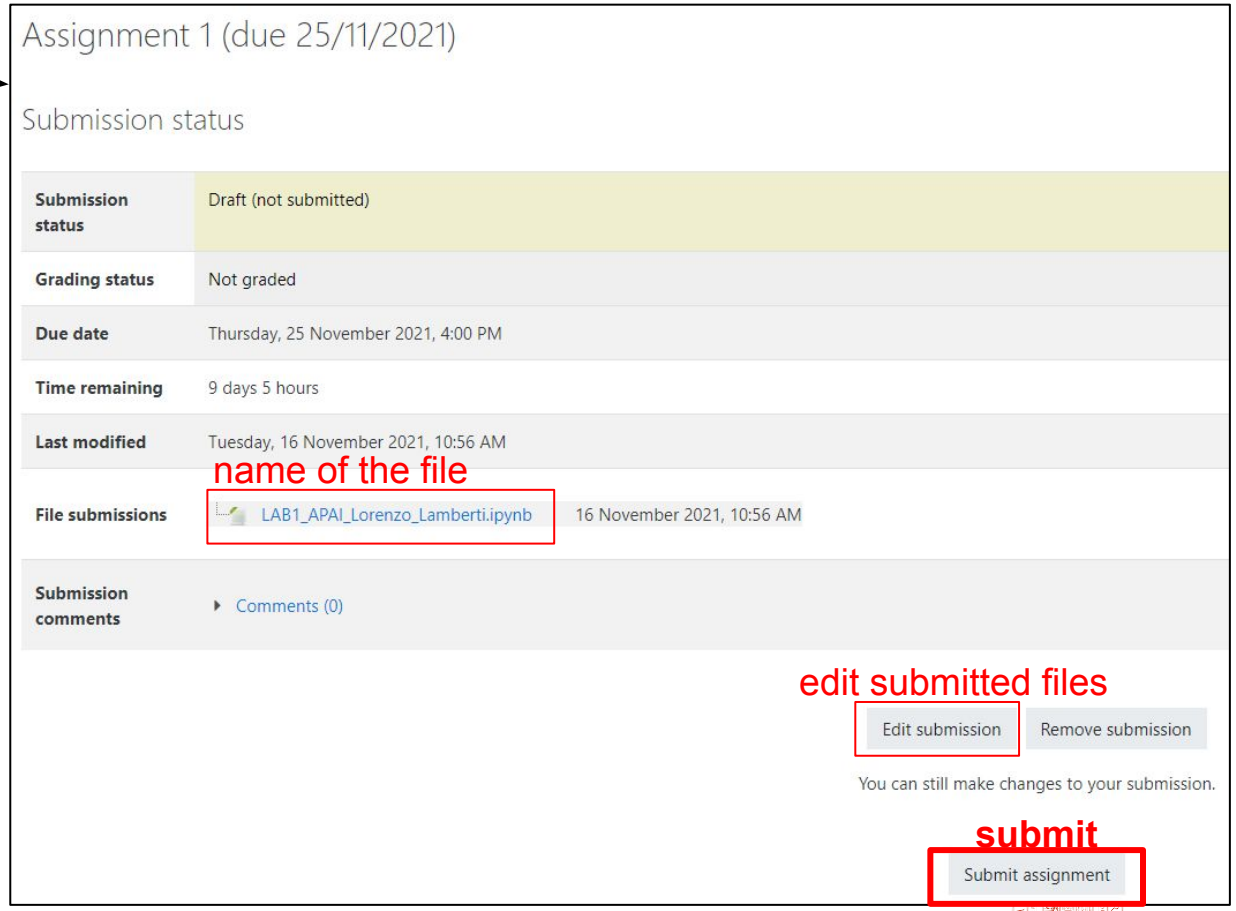
  LAB1_APAI_yourname.ipynb



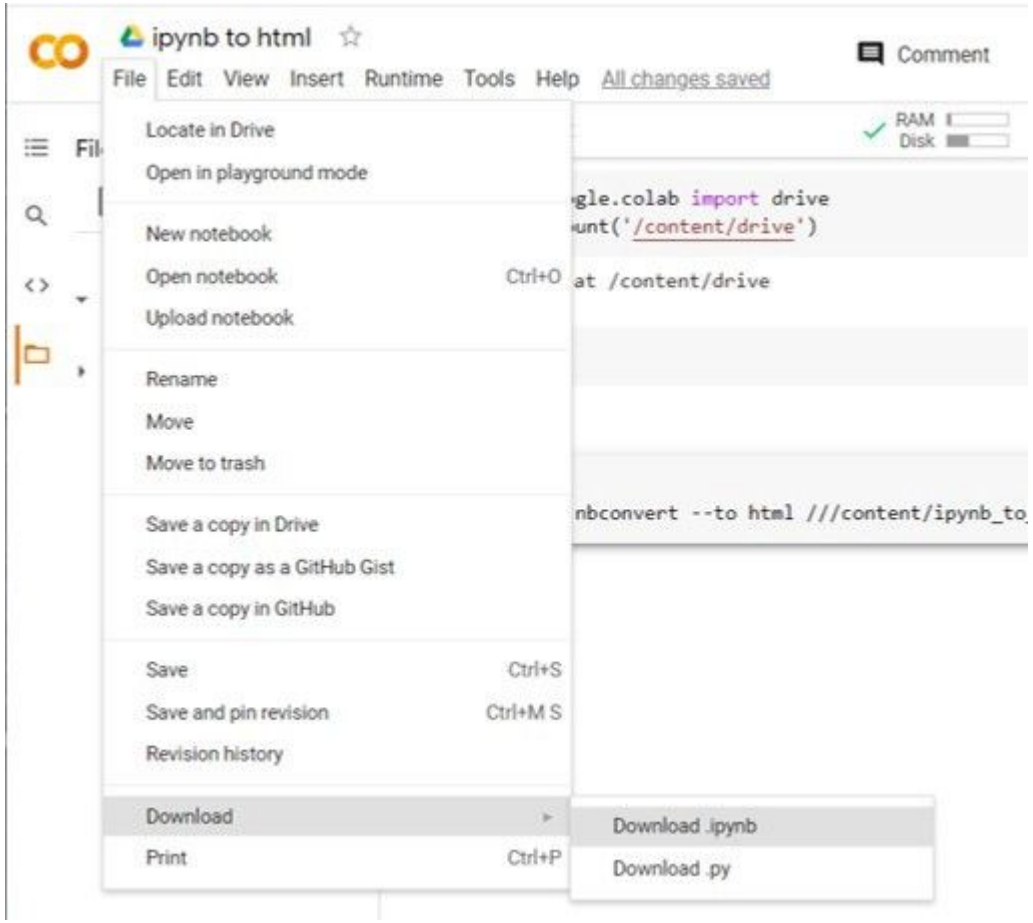
Important: the notebook must be pre-run by you. Outputs must be correct and visible when you download it.

LAB3 DEADLINE:

23/10/2025

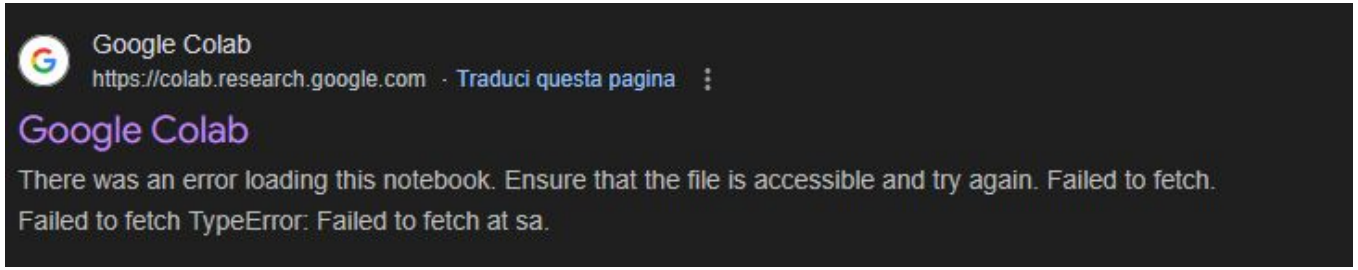


How to download the .ipynb file



Setup

1



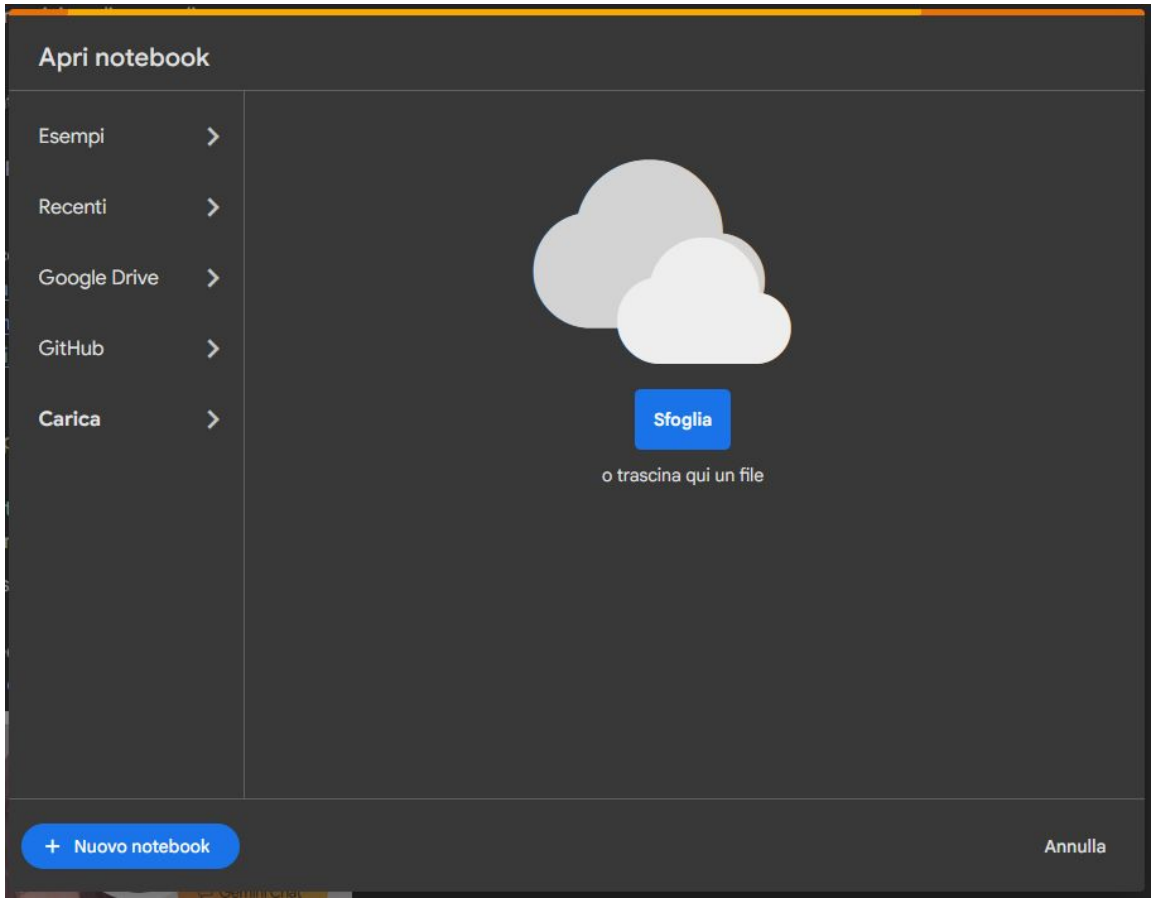
Seach for and open Google Colab (https://colab.research.google.com/)

2

Download the .ipynb file from today's repository:
https://github.com/EEESlab/APAI25-LAB03-DNN-Shrinking-and-Quantization

Setup

3



Upload the file to Google Colab (drag and drop or search)

3

Start working by CAREFULLY reading the setup and assignment text in EVERY block!!

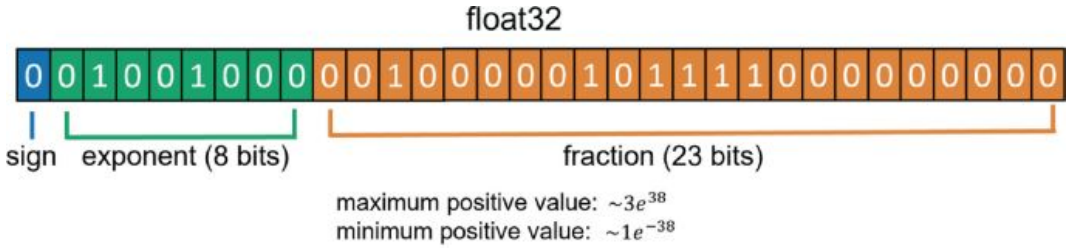


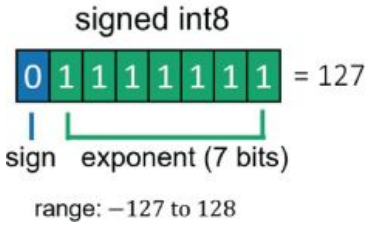
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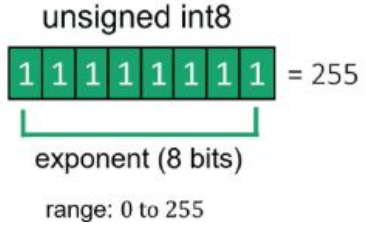
A bit of theory

Data Types and Quantization

IEEE 754 32-bit Floating Point format







Training time: FP32 for numerical accuracy! However, 1 element of activation / weights = 4 bytes (memory hungry for inference!)

Deployment: quantization! Smaller number of bits for a single element (e.g., 8 bits = 1 byte), but less numerical precision!!

Remember Lab1? **Data stored in 8 bits** (instead of 16 or 32) easily **overflow** or cannot represent **values outside range**! One possible solution: **quantization-aware training**!

Sources: https://en.wikipedia.org/wiki/Single-precision_floating-point_format
https://link.springer.com/chapter/10.1007/978-3-031-24538-1_1



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The LAB starts now !